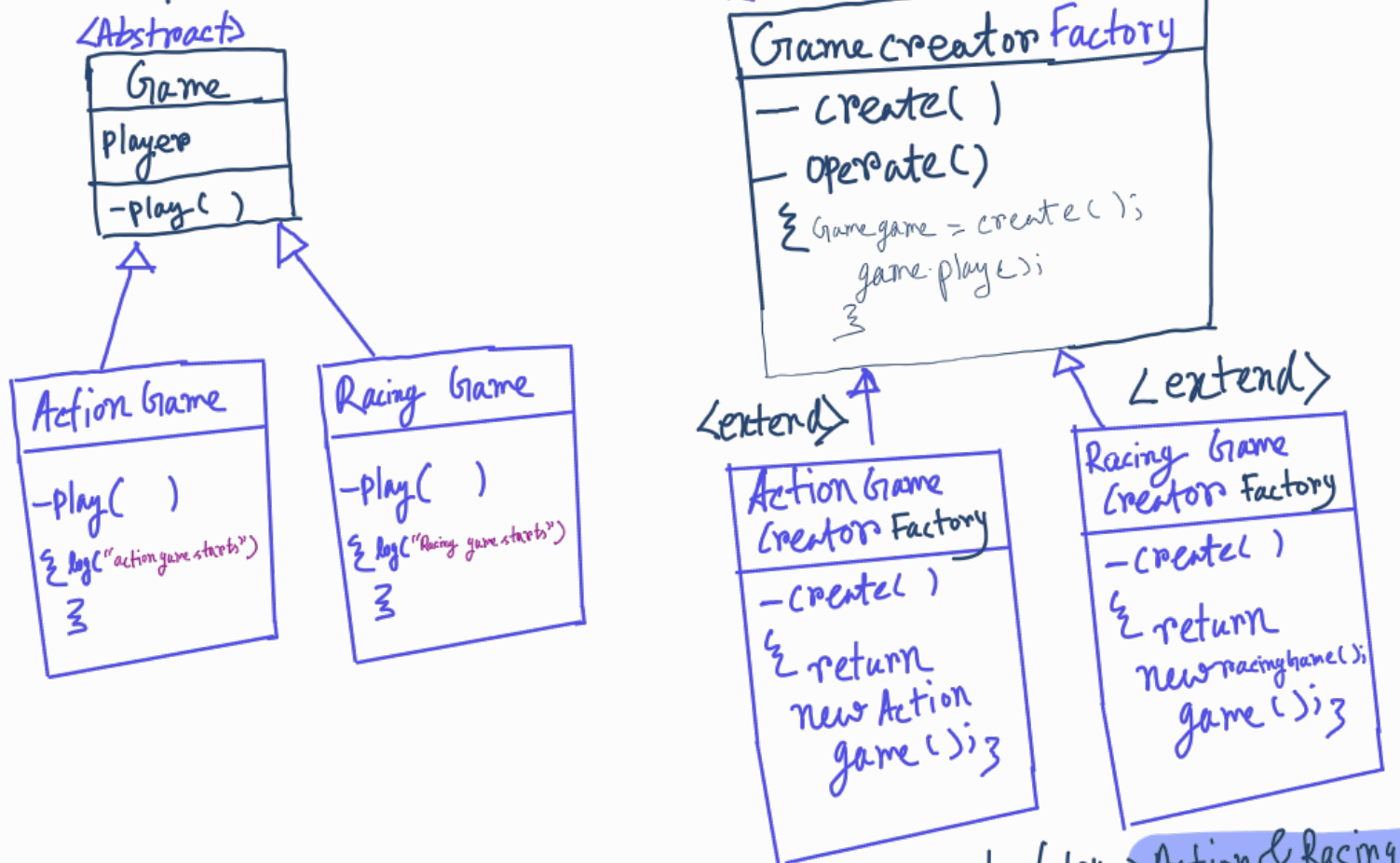


Factory Design Pattern: → Factory design pattern is a creational design pattern we use to create an object of a particular class.

Defination: — Factory design pattern is a design pattern that provides interface, for creating an object, but subclass to decide which class to instantiate.

Example: →



** Here subclass of Abstract class **game creator factory** → **Action & Racing game factory** decides which class to instantiate

When to use Factory Method-

- when we have to create objects of a particular class hierarchy.

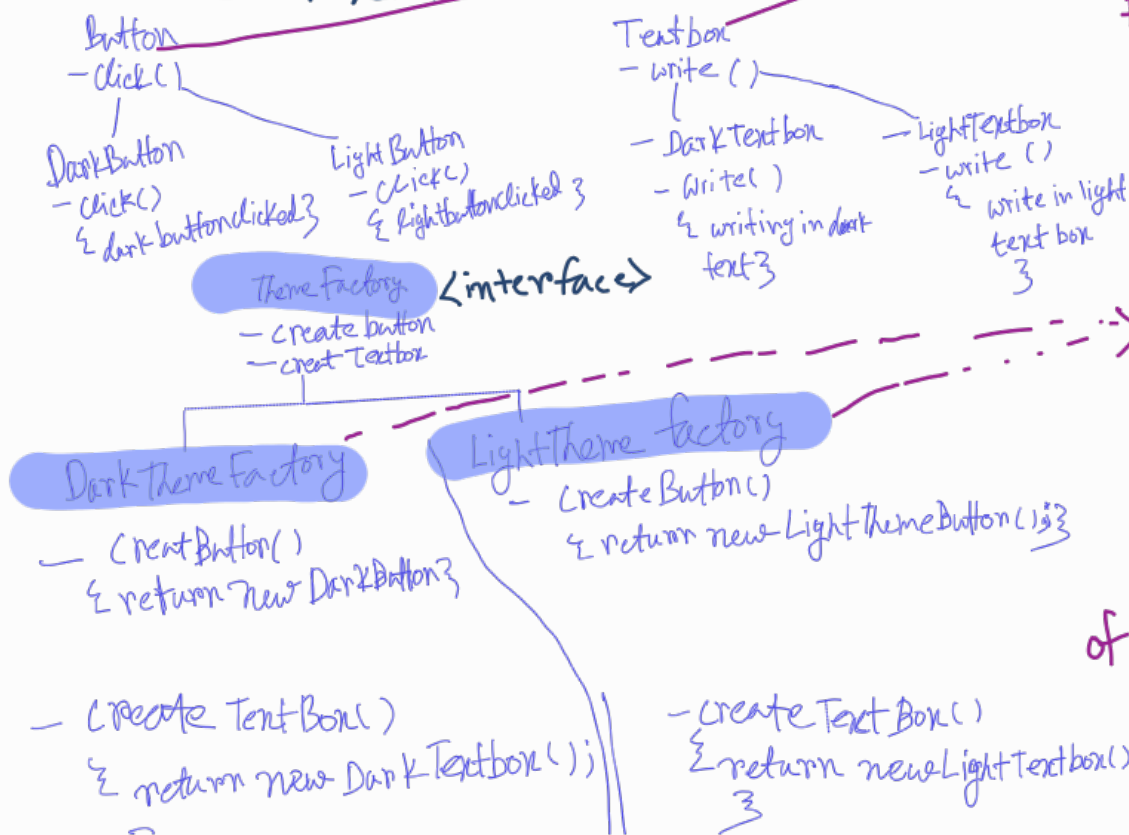
Abstract Factory design pattern:-

Defination:- → Abstract factory design pattern provides an interface to create families of related or depended objects.

Example:- For creating an UI we need-

- buttons
- text boxes.

Related
Classes
for the context
of UI



Concrete
Classes
which creates
family of objects
of related classes

When to use Abstract factory -
when we need to create families
of objects related or depended
classes. At those cases Abstract
factory provides safe interfaces, that gives related objects.
* When factory is responsible for
creating instances of multiple
co-related classes. not a single
class's implementations,

* When we need to make objects
that compatible with each other and
we have to be consistent, mixing up
will be fatal, then we use Abstract
factory which makes sure at the time
of creation, no mixup happens

Only Co-related class's object can be instantiated.

For example:- Dark theme UI Factory
only instantiate only dark-Themed
components.